

Circular Equal-Mass Hard Rods after Rotation Reduction: Exact Free-Flow Conjugacy, All Events, and Return Times

Route-A source-local certificate HCS-C296

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Abstract

For N indistinguishable equal-mass rods of length a on a circle of circumference $\ell > Na$, we identify the exact dynamical object on which the usual excluded-volume compression is globally valid. The conjugacy resolves binary, disjoint simultaneous, and genuine multiple collisions, preserves momentum and energy, and proves an explicit compact-interval no-Zeno bound. It also exposes a necessary boundary: the unreduced physical angle is an omitted cocycle, as the one-rod return period already demonstrates. We derive a necessary and sufficient return equation with one common translation and reduce it to an exact generalized-CRT criterion for the translation stabilizers of repeated-velocity position multisets. This gives minimal periods, symmetry shortening, fixed-shape and irrational boundaries, and a reproducible exact certificate without making an arithmetic or spectral target claim.

1 The corrected phase space

The frozen release obstruction identifier is **HEN-0280**. Write $\mathbb{T}_r = \mathbb{R}/r\mathbb{Z}$ and orient it clockwise. A rod is represented by its left endpoint. Choose a cyclic lift $x_1, \dots, x_N \in \mathbb{R}$ satisfying

$$x_{i+1} - x_i \geq a \quad (1 \leq i < N), \quad x_1 + \ell - x_N \geq a. \quad (1)$$

The free gaps are the left sides of (1) minus a ; they are nonnegative and sum to

$$L = \ell - Na > 0.$$

We quotient global constant rotations $x_i \mapsto x_i + r$ and cyclic changes of the first rod. On a contact face, incoming and outgoing velocity labels are glued by the equal-mass elastic reflection. At an intersection of faces, the gluing is generated by the adjacent transpositions in the maximal contact block. The resulting rotation-reduced, unlabeled, collision-glued phase space is denoted by $\mathcal{R}_{N,\ell,a}$.

The free target is

$$\mathcal{F}_{N,L} = ((\mathbb{T}_L \times \mathbb{R})^N) / (S_N \times \mathbb{T}_L), \quad (2)$$

where S_N permutes position-velocity pairs and $c \in \mathbb{T}_L$ sends every y_i to $y_i + c$ but leaves every velocity unchanged. Thus common drift is retained as phase data even though its spatial motion disappears in shape space. No time-dependent Galilean quotient is taken.

2 Compression and complete-flow conjugacy

For a representative satisfying (1), define

$$y_i = x_i - (i-1)a, \quad 1 \leq i \leq N, \quad (3)$$

and reduce the y_i modulo L in (2).

Lemma 1 (Cyclic-start identity). *If the next rod is chosen as the first rod, then the compressed lifted list is*

$$(y_2 + a, \dots, y_N + a, y_1 + L + a).$$

Consequently its class in (2) is unchanged.

Proof. The new lifted endpoint list is $(x_2, \dots, x_N, x_1 + \ell)$. Subtracting $(i - 1)a$ gives the displayed list, because $\ell - (N - 1)a = L + a$. Modulo L this is a cyclic permutation followed by one common translation by a . A global rod rotation similarly becomes a common translation. Hence (3) is well defined precisely on the reduced source. $\square$

Theorem 1 (Exact shape conjugacy). *Compression induces a bijection*

$$C : \mathcal{R}_{N,\ell,a} \longrightarrow \mathcal{F}_{N,L}.$$

For every $t \in \mathbb{R}$, it conjugates the complete hard-rod flow R_t to the free flow

$$F_t[(y_i, v_i)_{i=1}^N] = [(y_i + tv_i, v_i)_{i=1}^N], \quad CR_t = F_t C. \quad (4)$$

Proof. In cyclic order, (3) converts the free rod gaps exactly into the cyclic gaps between the y_i ; both are nonnegative and sum to L . Conversely, cyclically order any representative of the free points, choose lifts $y_1 \leq \dots \leq y_N \leq y_1 + L$, and put $x_i = y_i + (i - 1)a$. Then (1) holds. Common translation changes only the removed rod rotation, a different first point gives the cyclic relation in Lemma 1, and a permutation changes no unlabeled state. Ties are exactly contact faces, where the collision gluing removes the remaining velocity-order ambiguity. This proves bijectivity.

Between contacts, subtracting constants in (3) does not change velocities, so both sides follow (4). At an ordinary equal-mass collision, the rods exchange velocities; letting the two free phase points cross produces the same unlabeled pair. Adjacent transpositions generate every permutation within a simultaneous contact block. Therefore the identity persists through every collision, and the free group defines both forward and backward hard-rod evolution without an event-order convention. $\square$

Jepsen's equal-mass hard-rod dynamics and free-crossing mechanism are classical [1], as is the excluded-volume setting of the Tonks gas [2]. Our result is a normalized source-model closure and not a claim of literature priority.

3 All events, conservation, and no Zeno

Fix a free representative. For $i \neq j$ with $v_i \neq v_j$, its pair coincidences are exactly

$$E_{ij} = \{t \in \mathbb{R} : (y_i - y_j) + t(v_i - v_j) \in L\mathbb{Z}\}. \quad (5)$$

At one event time, group all indices with the same compressed position. A group with at least two distinct velocities is a maximal collision block; different positions give simultaneous disjoint blocks.

Theorem 2 (All-event law). *Immediately before a collision block, its spatial velocity list is decreasing up to ties; immediately after, it is increasing. Thus the event replaces the incoming list by the same multiset in sorted order. Binary collision is velocity exchange, and a genuine multiple collision is independent of any binary resolution order in the collision-glued quotient.*

Every event conserves

$$P = \sum_{i=1}^N v_i, \quad E = \frac{1}{2} \sum_{i=1}^N v_i^2.$$

For a horizon $H \geq 0$, each unequal-velocity pair has at most

$$\left\lfloor \frac{H|v_i - v_j|}{L} \right\rfloor + 2 \quad (6)$$

coincidences in $[0, H]$. Hence only finitely many aggregated events occur on every compact interval: the flow has no Zeno accumulation.

Proof. Near a common free position at time t_0 , the local positions are $y_0 + (t - t_0)v_i$. For $t < t_0$, increasing spatial order therefore reads the velocities from large to small; for $t > t_0$, it reads them from small to large. Sorting is a permutation, so its first and second power sums, hence P and E , are unchanged.

Equation (5) is one affine congruence. As t ranges over $[0, H]$, its left side crosses an interval of length $H|v_i - v_j|$; the number of multiples of L in that interval is bounded by (6). Summing over the finitely many pairs proves local finiteness. If $v_i = v_j$, the two free points either never coincide or remain coincident; the latter is a persistent contact and not an isolated event. $\square$

Initial contacts are allowed because the phase space glues both one-sided velocity assignments. The construction is reversible for all real time. The strict inequality $L > 0$ is indispensable; at close packing the target circle collapses.

4 Why the unreduced statement is false

Lemma 1 shows the obstruction directly: changing the cyclic starting rod translates all compressed points by a . A permutation quotient alone does not remove that translation. There is an even shorter dynamical counterexample. For $N = 1$ and $v \neq 0$, the full physical rod returns after

$$T_{\text{physical}} = \frac{\ell}{|v|}, \quad (7)$$

whereas a naive point on the free circle $L = \ell - a$ would return after $L/|v|$. After rotation reduction, the configurational shape is one point and every fixed-velocity phase state is an equilibrium, exactly as Theorem 1 says.

Thus the complete physical rotation angle is an additional cocycle. This paper neither reconstructs it nor claims a conjugacy of the unreduced phase space. The obstruction is part of the theorem boundary, not a removable choice of notation.

5 Complete return criterion

A return in (2) may use both defining symmetries. Consequently a state returns at $T > 0$ if and only if there exist $\sigma \in S_N$ and $c \in \mathbb{T}_L$ such that

$$y_i + Tv_i \equiv y_{\sigma(i)} + c \pmod{L}, \quad v_i = v_{\sigma(i)}, \quad 1 \leq i \leq N. \quad (8)$$

The common c is mandatory: deleting it would restore the angle that was removed from the source.

For a velocity value u , let

$$Y_u = \{y_i : v_i = u\}$$

be a multiset in $\mathbb{T}_L$ and define

$$H_u = \text{Stab}(Y_u) = \{h \in \mathbb{T}_L : Y_u + h = Y_u\}.$$

It is a finite cyclic subgroup. Write its order as d_u ; then

$$H_u = \left\{ \frac{kL}{d_u} : 0 \leq k < d_u \right\}, \quad d_u \mid |Y_u|. \quad (9)$$

Theorem 3 (Stabilizer and minimal-period theorem). *For $T > 0$, the following are equivalent:*

1. *the reduced hard-rod state returns at T ;*
2. *the equation (8) holds for some σ, c ;*
3. *the cosets $Tu + H_u$, over all distinct velocities u , have nonempty common intersection;*
4. *for every pair of distinct velocity values u, w ,*

$$\text{lcm}(d_u, d_w)T(u - w) \in LZ. \quad (10)$$

Let

$$\mathcal{W} = \{\text{lcm}(d_u, d_w)(u - w) : u < w\}.$$

If there is only one velocity value, the shape state is fixed and has no least positive period. Otherwise a positive return exists exactly when the nonzero elements of $\mathcal{W}$ are commensurable. If $\mathcal{W} \subset \omega\mathbb{Z}$ with primitive integer coefficients, then

$$T_{\min} = \frac{L}{\omega}. \quad (11)$$

Proof. The equivalence of the first two statements is the conjugacy and the definition of the quotient. Since σ must preserve velocity values, equation (8) restricted to one class is the multiset identity $Y_u + Tu = Y_u + c$, or $c \in Tu + H_u$. This proves the third statement.

Two such cosets intersect precisely when $T(u - w) \in H_u + H_w$. By (9), their sum is $H_{\text{lcm}(d_u, d_w)}$, giving (10). Pairwise compatibility is sufficient: choose a common multiple of all d_u and identify the finite subgroups with subgroups of one finite cyclic group; the assertion becomes the generalized Chinese remainder theorem for congruence classes. This proves the fourth equivalence.

Condition (10) is $Tz/L \in \mathbb{Z}$ for every $z \in \mathcal{W}$. Such a T exists exactly when the finitely many nonzero z lie in one discrete cyclic subgroup of $\mathbb{R}$. Taking its primitive positive generator ω gives (11); Bezout's identity for the primitive integer coefficients proves that no smaller positive T works. $\square$

If all velocities are pairwise distinct, every Y_u is a singleton and $d_u = 1$. Hence return reduces to

$$T(v_i - v_j) \in L\mathbb{Z} \quad \text{for all } i, j. \quad (12)$$

In particular, $(0, 1, \sqrt{2})$ has no positive return: (12) would make both T/L and $\sqrt{2}T/L$ integers. With repeated velocities, the factors d_u can shorten a period. The exact certificate includes two velocity classes on $L = 12$, each with order-two stabilizer; velocities ± 1 then have reduced minimal period 3, rather than the generic value 6.

6 Exact evidence and Route-A boundary

Seven rational scenarios audit 149 cells: 20 phase points, 23 pair crossings, 16 aggregated event times with 21 collision groups, 29 quotient-shape queries, 16 velocity-class stabilizers, seven return witnesses, one symbolic aperiodic control, seven conservation/no-Zeno rows, and nine boundary rows. The independent checker reconstructs every grid without importing producer code and reports 2,685 assertions. SymPy and exact finite cyclic groups check 159,064 identities; two isolated evidence builds are byte identical; and 96/96 hostile mutations are rejected, including direct evidence/YAML attacks on HEN-0280, duplicate top/nested JSON and YAML keys, a false unreduced conjugacy, altered event invariants, period witnesses, scope flags, and Route-B promotion.

The conservative Route-A result is

(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION),

with overall ROUTE_A_REJECTED. A1 is weak because the exact periodic-return structure has no intrinsic prime or prime-power bridge. A4 records the natural hard-core kinetic/Friedrichs quantization on the reduced collision chamber; no spectrum is analyzed here. Route B is disabled.

Under NO_BAD_EULER_OR_ROOT_NUMBER, nothing in this package is target arithmetic local data, a target Euler factor, a root number, an automorphy statement, a target divisor or counting law, a target functional equation, a target zero match, or a Hilbert–Pólya operator. There are no human or animal subjects and no personal data, so ethics approval is not applicable. No external funding or conflict of interest is reported. CRediT roles are conceptualization, formal analysis, software, validation, writing, and reproducibility curation. AI assistance was used in drafting and code generation; claims were checked against the proofs, strict schemas, independent reconstruction, symbolic identities, and hostile tests retained in the package.

References

- [1] D. W. Jepsen, “Dynamics of a Simple Many-Body System of Hard Rods,” *Journal of Mathematical Physics* 6(3) (1965), 405–413, [doi:10.1063/1.1704288](https://doi.org/10.1063/1.1704288).
- [2] L. Tonks, “The Complete Equation of State of One, Two and Three-Dimensional Gases of Hard Elastic Spheres,” *Physical Review* 50 (1936), 955–963, [doi:10.1103/PhysRev.50.955](https://doi.org/10.1103/PhysRev.50.955).